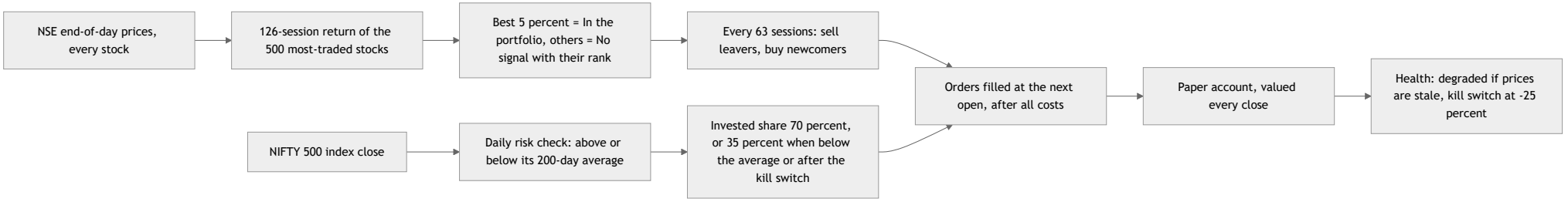
How ALADIN works (model E, version aladin-e-1)

Rewritten 2026-11-10 for the rebuilt ALADIN. The earlier models are described further down under "Before the rebuild"; at the owner's request they keep running and learning every night, and move behind the ALADIN BOT access code in Phase 6.

The big picture



Walkthrough. Every evening the Daily update saves the day's NSE prices. Model E then (1) fills any orders queued the evening before at today's opening prices, after costs; (2) values the paper account at today's close; (3) checks the data is fresh and the account is within its limits; (4) ranks the 500 most-traded stocks by their 6-month return; (5) every 63 sessions queues orders to sell stocks that left the best 5% and buy the new ones; and (6) if the NIFTY 500 crossed its 200-day average, or the kill switch fired, queues orders to resize every position. Nothing is filled at the close that produced the signal.

Data

Source	Used for	Updates	When missing
NSE end-of-day files + split-adjusted daily candles (data/terminal)	the 6-month return, traded value, the fills at the open, the account's value	every evening	prices more than 2 sessions old: **degraded**, no new decisions, signals hidden
NIFTY 500 index closes (data/terminal/idx)	the daily risk check	every evening	risk check not measured; no change of invested share

Nothing is filled with zeros or guesses. A stock whose price history has a one-day move above 45% (beyond any NSE price band) is left out as a likely unadjusted split or demerger.

Inputs

Name	Meaning	Formula in words	Look-back
r126	6-month return	close today ÷ close 126 sessions ago − 1	126 sessions
adv60	how much it trades	average of close × volume	60 sessions
rank	place among the 500	1 = highest r126	–
NIFTY 500 vs 200-day average	market trend	index close above or below the average of its last 200 closes	200 sessions

The model

- **Type:** a fixed ranking rule (no machine learning, nothing to overfit beyond three settings chosen in Phase 4).
- **What it expects:** the stocks that rose most over 6 months, as a group, to beat the average stock over the next quarter. It does not predict any single stock; most single stocks in the best group trail the average, a minority win big.
- **Settings and where they came from:** 126 sessions, best 5%, every 63 sessions, 70% invested (35% when risk is off), all chosen in the Phase 4 walk-forward test and approved by the owner. Kill switch −25% (the owner's limit). Position cap 5% (a quarter of the Kelly fraction implied by the measured edge is 8%, so 5% binds).

The measured chance (shown with every stock)

From 2010–2026 signal dates, every 5 sessions, outcomes over the next 63 sessions, before costs, companies listed today only:

Rank group	Beat the average stock	Average lead (95% range)	Typical (median) lead	Closed higher
Best 5% (in the portfolio)	46.6%	+2.55% (+1.94 to +3.17)	−2.0%	52.3%
Next 5%	47.7%	+1.89% (+1.45 to +2.36)	−1.1%	53.4%
80th–90th percentile	47.7%	+1.25%	−0.9%	54.4%
50th–80th	46.7%	+0.53%	−1.3%	54.8%
20th–50th	44.3%	−0.44%	−2.0%	52.9%
Worst 20%	40.1%	−1.92%	−3.6%	49.5%

What this means: the ranking is in the right order on average, but even the best group's typical stock trails the average; the edge comes from a minority of large winners. That is why E holds a basket of about 25 and never relies on one stock.

Three worked examples (6 November 2026 close)

Step	RELIANCE (large-cap)	PERSISTENT (mid-cap)	SAILIFE (near its 52-week high on heavy volume)
6-month return (r126)	−2.9%	+24.9%	+64.5%
Traded value (60-day average)	₹1,814 Cr a day	₹322 Cr	₹126 Cr
Rank among the 500	322 (better than 36%)	133 (better than 74%)	35 (better than 93%)
Cut-off for the portfolio	rank 25, a 6-month return of +76.7%	same	same
State	No signal	No signal	No signal (just outside the best 25)
Its group's measured result	20th–50th: beat the average stock 44.3% of the time, average −0.4% a quarter	50th–80th: 46.7%, +0.5%	next 5%: 47.7%, +1.9%
Market risk check	NIFTY 500 23,457 above its 200-day average 22,763: risk on, 70% invested	same	same

What this means: SAILIFE, which the old ALADIN showed red, now ranks 35th of 500, near the top; it is not bought only because the basket takes the best 25. RELIANCE sits in the lower middle. No colour says "buy" or "sell": each stock shows its rank, its state and its group's measured record.

Health

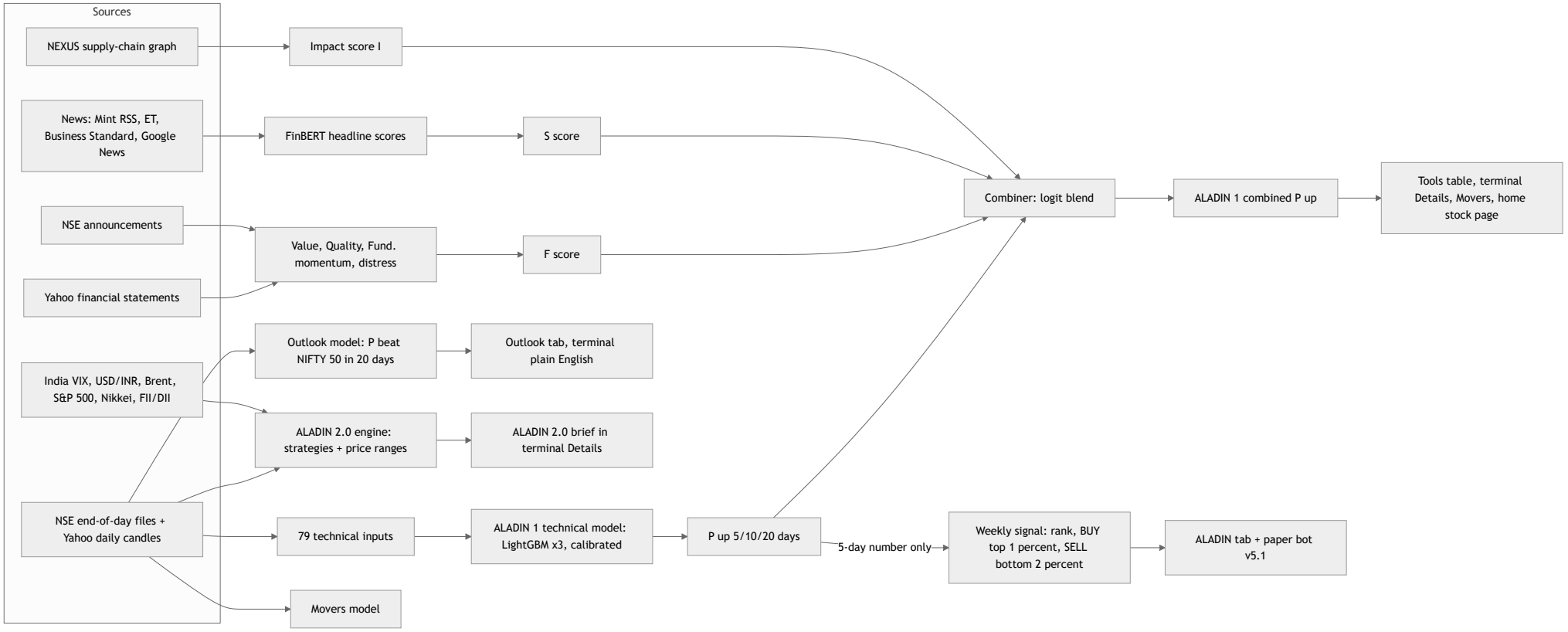
Runs in the Daily update (scripts/aladin_e.py); tests: tests/test_aladin_e.py (11, all passing: date alignment, next-open fills, running twice changes nothing, stale data, risk halving, kill switch, costs including DP, position cap, calibration order, no price target, data-error jumps). The paper account starts on its first run after the owner approves Phase 5.

Before the rebuild (still running, moving behind ALADIN BOT in Phase 6)

Written 2026-11-08 from the code on `main` and the data published for the 2026-11-06 close. Nothing was changed. Words in *italics* are explained in GLOSSARY.md.

1.1 The big picture

ALADIN is five prediction systems that share data. Only **ALADIN 1** blends the three fronts you know (technical, fundamental, sentiment). The **ALADIN tab** shows a different system (the weekly signal), which uses ALADIN 1's technical 5-day number and nothing else.



Walkthrough. Every evening after the market closes the *Daily update* job saves the day's prices and runs the Outlook model. When it ends it starts *ALADIN nightly*: it refreshes about 400 companies' statements, scores filings, computes the supply-chain impact, builds 79 technical inputs for every stock, runs the technical model for 5, 10 and 20 days, converts its raw score into a *calibrated* probability, and blends it with the fundamental and sentiment scores. Then *ALADIN2 nightly* starts: it runs the ALADIN 2.0 engine, writes the weekly BUY/SELL list from the technical 5-day number, and updates the paper bot. The sentiment job runs separately about every 30 minutes. The website reads the saved files; the colour you see is chosen in the browser by fixed cut-offs (section 1.5).

1.2 Data, source by source

Source	Gives	Fetched by	Updates	Usual age when used	When missing or late
NSE end-of-day files + Yahoo candles	daily open/high/low/close/volume, split-adjusted (not dividend-adjusted)	nse_eod.py, terminal_hist.py (Daily update)	every evening	same day's close	a stock whose last bar is >10 days old is not scored (aladin_model.py)
Yahoo statements	annual + quarterly income, balance sheet	aladin_fund.py, ~400 companies a night	rotating; each company every few nights	days to weeks; **filing dates not stored**	F is None, its weight drops out of the blend (counted as 0). RELIANCE has no F today
NSE announcements	text of filings, scored by FinBERT	aladin_filings.py	nightly, last 90 days, 15-minute budget	1 day	the NLP part of F is left out
News	headlines	aladin_sentiment_engine.py, news_wire.py	about every 30 min	minutes to hours; decays with a 24-hour half-life-like weight	S is None, counted as 0 in the blend (most stocks: only 2 headlines for RELIANCE, none for PERSISTENT, SAILIFE)
Market series (VIX, INR, Brent, S&P 500, Nikkei, US 10y)	context for ALADIN 2.0	ALADIN2 nightly	daily	0–1 day	the feature is empty that day
FII/DII flows	net buying	institutional.py	daily	1 day	last saved day kept; used inside sentiment as "flow"
NEXUS graph	supply-chain links	nexus.yml, graph_impact.py	weekly	days	I is None (true for almost every stock)
Liquidity sweeps	live price patterns	the owner's PC only	live	–	absent on GitHub, so always 0 in the nightly blend

Silent fills. In the blend, a missing F, S, I or sweep counts as **0** (no push either way). The technical model itself lets LightGBM handle missing inputs (it sends them down a learned branch). Nothing is filled with an old value except prices (last saved copy).

1.3 Inputs ("features")

Technical, 79 inputs in total (scripts/predict_model.py `features`, scripts/aladin_model.py `stock_frame`, `build_dataset`):

Group	Inputs (code name: meaning)	Look-back	Scaling
Returns	r1, r5, r10, r20, r60, r120, r250: log change in price over 1 day to 1 year; mom12_1: 1-year minus 1-month return	1–250 days	raw; several also ranked 0–1 across stocks that day (`_rk`)
Trend	d_sma20/50/200: price ÷ its 20/50/200-day average − 1; sma50_slope, sma200_slope: change of those averages over 10/20 days	20–220 days	raw
Oscillators	rsi2, rsi14 (relative strength 0–100), stoch14 (where the close sits in the 14-day range), cci_z, adx (trend strength), di_diff, macd_h, bb_pct, bb_w	2–26 days	raw
Volatility	atr_pct (typical daily range ÷ price), vol20, vol60, vol_ratio, skew60, range20, gap20	14–60 days	raw
Position	hi52, lo52 (distance from 52-week high/low), dd60 (fall from the 60-day high)	60–252 days	raw
Volume	vz20 (today's volume vs its 20-day average, log), vtrend, turn (traded value)	5–60 days	raw
Market link	beta, corr60, idio60 (stock-specific volatility), rel20, rel60 (return vs NIFTY 50), sec_rel60 (vs its sector)	20–250 days	raw
Market-wide	m_r5/20/60 (NIFTY returns), m_d200, m_vol20, m_br50/m_br200 (share of stocks above their 50/200-day average), m_disp	5–200 days	same for every stock that day
Quant extras	pca_z, pca_r2 (move not explained by market factors), coint_z, coint_hl, coint_p (gap to a paired stock), hmm_p_highvol (chance the market is in a turbulent state), jump_lam, jump_recent, jump_bv	20–250 days	raw
Ranks	17 of the above ranked 0 (lowest) to 1 (highest) among that day's liquid stocks	–	percentile
Labels	ind: industry (category)	–	category
Calendar	**moy: month of the year (1–12); dom: day of the month (1–31)**	–	raw. Added because it "improved 8 of 13 test years"

Fundamental (aladin_fund.py): value, quality, fundamental momentum (each a z-score, compared across companies), NLP tone of filings, distress band. `F = 100 × tanh(z ÷ 1.5)` where `z = 0.35 value + 0.35 quality + 0.20 momentum

- 0.10 NLP − distress penalty` (weights re-scaled over the parts present). Range −100 to +100.

Sentiment (aladin_sentiment_engine.py): each headline gets FinBERT's P(positive) − P(negative); weighted by age (`exp(-hours ÷ 24)`) and source; per stock `raw = 0.60 news + 0.15 retail + 0.15 FII/DII flow` (re-scaled over the parts present) **plus a hand-set geopolitical adjustment** (`geo_adjust` , from geo_exposure.json); `S`

`= 100 × raw`, clipped to ±100. Levels: BAD ≤ −60, POOR ≤ −20, NEUTRAL, GOOD ≥ 20, EXCELLENT ≥ 60.

1.4 Each model

	ALADIN 1 technical	Outlook	ALADIN 2.0 engine	Weekly signal	Movers
Type	machine learning: 3 LightGBM tree models (different random seeds), averaged; for 10 days a calm/stressed blend	machine learning: 3 LightGBM classifiers + 1 regressor, rank-averaged	strategy library (trend, reversal, volatility, relative strength, delivery), strategy life cycle, volatility model for ranges	a rule on ALADIN 1's number	machine learning
Predicts ("target")	close in 5/10/20 trading days > today's close (raw direction, **no costs**, not vs the market)	20-day return beats NIFTY 50's	per strategy: net return after costs; ranges: price quantiles	next 5 days, enter next open	5-day move size and direction
Trained on	2008/2009 onward, main-board stocks with ≥ ₹5 Cr a day **on that day** (point in time); walk-forward, test years 2013–2026 with a 15-day gap; final model retrained nightly on everything	NIFTY 500 **as of today** (today's list), 2008 onward, walk-forward 2013–2026, 25-day gap	2005 onward, test from 2012-07, 63-day blocks, 20-day purge + 5-day embargo	the weekly study 2013–2026 (310,816 stock-weeks)	2016–2026
Outputs	calibrated probability 2–98%	calibrated probability, decile	product state; forecast range; signal only if Validated	BUY / SELL / HOLD, a stop, a "target"	size + direction
Thresholds and where from	calibration: isotonic map fitted on the 2013–2026 unseen-year record; confidence Low < 3 pts from 50%, Medium < 7, else High	none for the colour	Validated: ≥ 60 unseen trades, ≥ 60 live days, calibration error ≤ 5%, net ≥ 5 bps; kill switch at calibration error > 5%	BUY ≥ 99th percentile, SELL ≤ 2nd (tested: chosen on 2013–2020, held on 2021–2026); stop = 2 × ATR	pre-declared gates
Weight in a blend	the base of the ALADIN 1 blend	–	–	–	–
Measured skill (unseen data)	AUC 0.546 at 10 days (0.5 = chance); top tenth went up 54.3%, bottom tenth 43.5%	AUC 0.534	live: **suspended** (see 1.7)	BUY top 1%: +27 bps a week after costs in the study	direction AUC 0.511: **failed**

The ALADIN 1 blend (scripts/aladin_model.py `combine_py`, identical in desk-aladin.js): `logit(p) = logit(p_technical) + 0.20 × F_adj/100 + 0.12 × S/100 + 0.18 × sweep/100`, with `F_adj = F − I`. Weights 0.20 / 0.12 / 0.18 are **set by hand ("prior")**: a fitted blend is only tried after 120 trading days of matured live predictions (16 so far). Missing parts count as 0. Result clipped to 2–98%.

1.5 From a number to a colour

Page	Number shown	Green	Grey	Red
ALADIN tab, signal badge	weekly signal	BUY (rank ≥ 99th percentile)	HOLD	SELL (rank ≤ 2nd)
ALADIN tab, "stop loss" cell	distance of the stop from the price	stop above price	–	**stop below price: always, for every BUY**
ALADIN tab, "target" cell	distance of the range edge	above price	–	below price ("not measured" for 12 of 13 BUYs this week)
Terminal Details, Tools table (`probBar`), Movers	ALADIN 1 combined P(up), 10 days	≥ 55%	45–55%	≤ 45%
Home stock page, "What our models see"	ALADIN 1 technical P(up) at 5, 10, 20 days	**≥ 50%**	–	**< 50%**
Outlook tab, terminal	P(beat NIFTY 50)	≥ 55%	45–55%	≤ 45%
ALADIN 2.0 brief	product state	Validated + signal	Provisional / HOLD	–

Today's spread of ALADIN 1 (2026-11-06 close, 2,458 stocks).

Horizon	Lowest	Median	Highest	Stocks above 50%	Different values
5 days (technical)	45.3%	51.6%	52.4%	85.6%	17
10 days (technical)	41.0%	**44.5%**	**50.0%**	**0%**	21
20 days (technical)	45.9%	48.5%	53.8%	16.3%	19
10 days (combined)	–	44.5%	–	0.3%	–

What this means: on every page that colours ALADIN 1's 10-day number, no stock can be green, and on the home stock page every stock's 10-day bar is red. The 13 BUY stocks on the ALADIN tab all have the same 5-day value, 52.4%.

1.6 Three worked examples (2026-11-06 close)

RELIANCE (large-cap)

Step	Numbers	What happened
Raw data	close ₹1,320; +6.8% vs 50-day average, −0.4% vs 200-day; 1 month +9.3%, 1 year −11.3%; 82% of its 52-week high; volume 1.09× normal	a recovering large-cap, flat on the year
Technical model	P(up) 5d 51.6%, **10d 44.3%**, 20d 49.5%	the 10-day number is low
Biggest reasons (10-day, in log-odds)	**day of month −0.245**, industry +0.042, month of year +0.039, 1-week return rank −0.024, RSI(2) −0.020	the calendar alone outweighs everything about the company
Fundamental	F = **none** (no statements in the data)	counted as 0
Sentiment	S = +28.9 (GOOD, 2 headlines)	small push up
Blend, 10 days	logit(0.443) = −0.229; + 0.12 × 28.9/100 = +0.035; total −0.194 → **45.2%** (published 45.05%)	blend reproduced by hand
Colour	Terminal: grey (45–55%). Home stock page: 10-day bar **red**. Weekly rank 50th percentile: HOLD	

PERSISTENT (mid-cap)

Step	Numbers	What happened
Raw data	close ₹6,053.5; +7.1% vs 50-day, +13.8% vs 200-day; 1 month +9.9%, 3 months +10.9%; 92% of its 52-week high	in an uptrend
Technical model	5d 52.4%, **10d 46.8%**, 20d 52.7%	
Biggest reasons	**day of month −0.240**, industry +0.139, month +0.030, 1-week return rank +0.019, 52-week-high rank +0.013	calendar again the largest
Fundamental	value −0.87, quality +0.06, momentum +0.17, distress Low → F = −21.8	expensive, so a push down
Sentiment	none	0
Blend, 10 days	−0.128 + 0.20 × (−21.8)/100 = −0.044 → −0.172 → **45.7%** (published 45.7%)	
Colour	Terminal: grey. Home stock page: 10-day bar **red**. Weekly rank 88th percentile: HOLD. Outlook 50.6% (decile 7)	

SAILIFE, Sai Life Sciences (should look bullish)

Step	Numbers	What happened
Raw data	close ₹1,785.1; **98.4% of its 52-week high**; +16.6% vs 50-day, +47.1% vs 200-day; 3 months +26%, 1 year +94%; **volume 11.8× normal**	a textbook breakout on heavy volume
Technical model	5d 51.6%, **10d 44.1%**, 20d 50.6%	reads as below a coin flip
Biggest reasons	**day of month −0.227**, industry +0.101, 1-week return rank −0.026, month +0.026, RSI(2) −0.020	the breakout is not among the top reasons
Fundamental	value −1.07, quality −0.28, momentum +0.99, distress Low → F = −8.2	
Sentiment	none	
Blend, 10 days	−0.237 + 0.20 × (−8.2)/100 = −0.016 → −0.254 → **43.7%** (published 43.65%)	
Colour	Terminal: **red** (≤ 45%). Home stock page: 10-day bar **red**. Weekly rank 39th percentile: HOLD. Outlook 47.6%	

1.7 Health snapshot (2026-11-08)

Item	State
ALADIN 1	probabilities as of 2026-11-06; models retrained every night; unseen-year record and calibration last rebuilt up to 2026-09-15 (10-day)
Outlook	as of 2026-11-06; record to 2026-10-08
ALADIN 2.0 engine	**all 1,382 stocks Suspended by its own kill switch**: live calibration error 14.7% (limit 5%); live 50% range caught the price 44.5% of the time, 80% range 75.2%. So no ALADIN 2.0 signal anywhere
Weekly signal	as of 2026-11-06: 13 BUY, 26 SELL, 1,245 HOLD; all "Provisional" (no live record yet)
Movers	failed its own test (direction AUC 0.511); its live list is not on `main`
Fitted blend	not tried yet: 16 of 120 matured days
Sweep and impact terms	not validated: both effectively 0
Jobs	last 6 runs of ALADIN nightly, ALADIN2 nightly and ALADIN sentiment all succeeded. ALADIN nightly ran 6 times in 15 hours on 7–8 Nov because Daily update ran 5 times (schedule, re-starts and a failed run); each Daily update starts the ALADIN chain. Harmless but wasteful

Item	State
Tests	288 ALADIN 2.0 + ALADIN Python tests pass; 34 of 35 ALADIN 1 model tests pass (the one failure is a timing test that depends on this PC's speed, known before); 33 browser-code tests pass
Data gaps	RELIANCE and others without statements (F none); news covers few stocks; filing dates not stored; past NIFTY 500 lists not available; the Outlook model trains on today's NIFTY 500 list